Space Diving

Team 649: Problem A

November 5, 2023

Abstract

Our skydiver (nicknamed *Akashchari* in the local language) wants to jump from space. So we tried in this paper to discover the safest maximum altitude for him to jump. We analyzed several characteristics such as the Karman Line, Stratospheric Jump, Parachute Opening Shock, and various re-entry situations to determine what maximum altitude is safest for him. We examined the atmosphere's levels, pressure, and temperature. We also studied and evaluated data for previous skydivers such as Felix Baumgartner and Joseph Kittinger. We received varied outcomes depending on the conditions. Finally, we settled on 89.8 kilometers as the proper answer.

Keywords

Re-entry — G-forces — Free fall.

1 Introduction and Problem Restatement

Total mass with 190 kg skydiver has jumped from n altitude. We need to find the safest maximum altitude for him to land.

2 Notations

Variables	Explanations	Value
V	velocity of free falling object	
g	free fall acceleration	$9.81m/s^2$
K	integral parameter introduced to simplify math transformations, can be considered	
m	mass of falling object	
190kg z	vertical coordinate of falling object	
C_d	drag coefficient in Earth's atmosphere at sea level	
t	time	
ρ	atmospheric density at sea level	
A	cross-sectional area of falling object	
λ	characteristic height for standard atmosphere (Laplace's isothermal atmosphere)	7.46 Km
RMSSD	Root Mean Square Successive Difference	
Ma	Mach Number	
Re	Reynold Number	

Table 1: Used Notations

3 Assumptions

We've assumed that our skydiver's body is cylindrical in this case. We also used data from the current NASA spacesuit and parachute. We also recorded the heart rate of a professional skydiver.

4 Analysis

4.1 Skydiving Basics

Sky diving is an extreme sport that involves leaping out of a moving airplane, helicopter, gondola, or other vehicle at high heights, often between 10,000 and 14,000 feet (3-4.2 kilometers. and up to 25000 feet (7.5 km) for military (up to 30000 ft (9 km) for some special missions. Exit and speed gain, free fall at terminal velocity, parachute opening, parachute flying, and landing are the five separate stages of sky diving. The jumper exits the airplane and proceeds to accelerate downward until he achieves terminal velocity. "You will accelerate for twelve seconds until you reach about 120 mph downward, which is nominal terminal velocity, that speed at which the pull of gravity (force on your body) equals your wind resistance". [Poynter, Turoff, 2007, p.10]. The average human body in the appropriate skydiving position (on the belly, knees, and arms half bent, back arched) will attain typical skydiving terminal velocity from a distance of up to 8 km. Due to thin air

and less drag, jumps from heights more than 8 km will result in terminal velocity at substantially higher speeds. Skydivers normally open their parachute between 2000 and 5000 feet (height varies on experience). This research looks at openings that provide anything from 1-2 to up to 18g of shock to the body and endure anywhere from 2 to 8 seconds. The parachutist guides the canopy in the desired direction, modifies its speed as needed, and seeks for a location to land [Poynter, Turoff, 2007]. A smooth landing keeps the skydiver on his feet [Poynter, Turoff, 2007]. Regular skydiving, sport skydiving, free-flying, base jumping, wingsuit flying, high altitude (space jumping), and military free fall are the **seven primary types of skydiving**.

4.2 Earth's Atmosphere & Kármán Line

The Earth's atmosphere is divided into five major layers and several secondary levels. The principal layers are, from lowest to highest, the troposphere, stratosphere, mesosphere, thermosphere, and exosphere.

We live in the *trposhepre*, the layer of the Earth's atmosphere closest to the surface and the densest of all. The troposphere holds the majority of the mass of the Earth's atmosphere, with various sources estimating that it includes 75% to 85% of the total mass of the atmosphere. Tropo is Greek for "change" or "turn." This layer's conditions are constantly changing as gas molecules rise and fall. In meteorology, skydiving, and aviation, there is a "rule of thumb" that states that temperature drops by 2 degrees Celsius for every 1000 feet of height.

Strato is an Italian word that means "layer." Because gases do not mix much from one layer to the next, stratospheric layers are more stable. Consider the ozone layer to help you recall that temperatures in this layer rise with altitude. Three oxygen atoms make up an ozone molecule. Heat is produced when three oxygen molecules come together. Furthermore, the ozone molecules filter the sun's harmful UV energy.

The gases in the *Mesosphere*, including oxygen molecules, continue to thin with increasing height. As a result, the influence of ultraviolet light warming grows less and less, resulting in a reduction in temperature with height. At the mesopause (the boundary between the mesosphere and the thermosphere), the temperature drops from roughly -15°C (5°F) to as low as -120°C (-184°F). The gases in the mesosphere, on the other hand, are still dense enough to slow meteors when they enter the atmosphere and burn up, creating fiery trails in the night sky. The intermediate atmosphere includes both the stratosphere and the mesosphere.

Although the thermosphere is considered part of the Earth's atmosphere, the air density in this layer is so low that the majority of the thermosphere is what we ordinarily think of as outer space. In actuality, the most frequent definition is that space begins at a height of 100 km (62 miles), at the bottom of the thermosphere.

As known Exosphere objects get heated when exposed to direct sunlight. Because the sun is so brilliant up there, items in the sun heat up rapidly. However, items in the shadow might become quite chilly. For example, one side of a satellite (the sunny side) may be extremely hot, while the other side (in the shadow) may be quite cold.

The Kármán line is a barrier between the Earth's atmosphere and space that is 62 miles (100 kilometers) above mean sea level. However, determining where space begins may be difficult and depends on who you ask. This is because the Earth's atmosphere does not stop abruptly but instead becomes thinner and thinner at greater altitudes, implying that there is no clear upper boundary.

4.3 Effect of Skydiving on the Human Body

We have gone through research to understand the effects of Skydiving on the Human Body. The study looks at how sensation-seeking (SS) qualities motivate people to seek novelty and risk in high-intensity contexts. It does, however, endeavor to comprehend the psychophysiological processes underpinning SS behavior and if novelty-seeking contributes to physiological arousal. They measured autonomic nervous system (ANS) activity in 44 skydivers before, during, and after their jumps with an ambulatory device. The study indicated that both novice and experienced jumpers had an increase in heart rate (HR) during the leap, showing that repeated exposure to danger did not result in habituation. Baseline functional variations across groups imply that novelty may initially inspire SS activities. Surprisingly, both the sympathetic and parasympathetic nervous systems coactivated throughout the leap, demonstrating that all ANS components are necessary to cope with extreme demands.

- Sympathetic nervous system activity
  RMSSD, as a level-1 variable, did not predict HR substantially ($\beta = .09, t = .19, p = .85$). This discovery is critical to our thesis that HR was indexing SNS activity. Because the data was centered on the moment of leap, the level-1 intercept represents the HR values at the time of jump. The preliminary interval did not substantially predict HR levels, according to the model ($\beta = 38.70, t = 1.16, p = .256$), hence HR remained lower than the peak at jump throughout this period. There was a trend toward the quadratic preliminary interval predicting HR ($\beta = 39.03, t = 1.75, p = .08$), indicating that HR was non-linear more than an hour before the jump. HR increased considerably throughout the anticipation interval, b = 12.86, t = 3.03, p =.005, and followed a linear trend. The jump interval captured the peak in HR levels ($\beta = 288.72, t = 8.19, p.0001$), and the quadratic transformation of this variable was also significant ($\beta = 1489.87, t = 13.45, p.0001$), indicating peak levels were attained during the relatively brief interval immediately before and after the jump. HR decreased further during the recovery time, $\beta = 67.47, t = 2.71, p.011$. HR changes across time accounted for 55% of the total variance in HR values.Between-individual variability was substantial throughout the preparatory interval (v2 (12) = 535.59, p.0001), anticipation ($\chi^2(12) = 488.76, p.0001$), jump ($\chi^2(12) = 1646.23, p.0001$), and recovery ($\chi^2(12) = 842.21, p.0001$). These data imply that each individual's pattern of HR activity was unique.

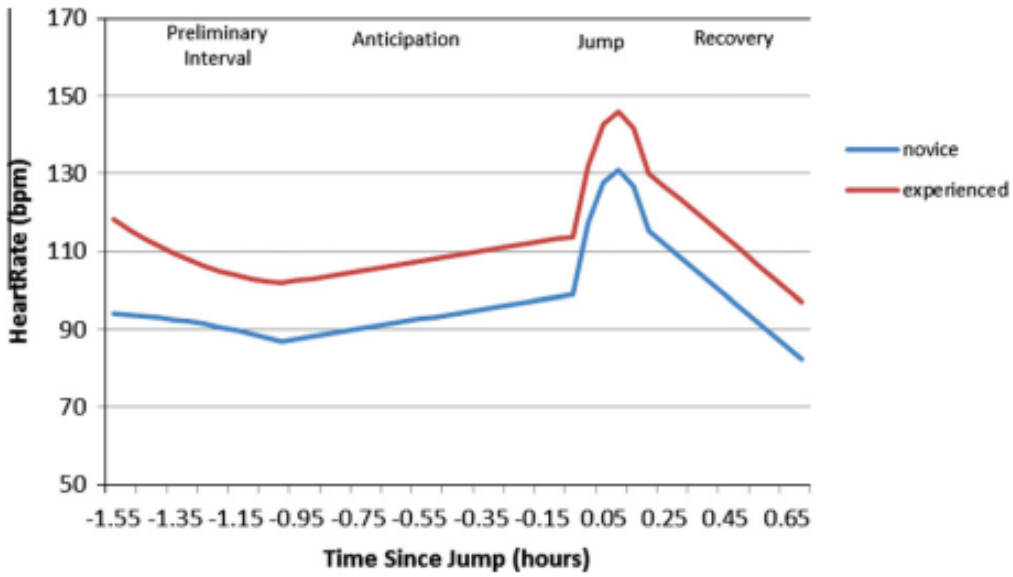

Figure 1: Before, during, and after skydiving, heart rates of novice and expert jumpers were measured.

- Parasympathetic nervous system activity
  The level-1 model found that the interval representing the preliminary interval did not predict RMSSD levels appreciably, $\beta = .083, t = .85, p = .339$. RMSSD decreased considerably for the majority of participants throughout the anticipation interval, $\beta = 1.21, t = 4.44, p.0001$, and the change seemed quadratic, $\beta = .853, t = 3.33, p.0001$. RMSSD peaked at the jump, $\beta = 5.22, t = 3.87, p.0001$, and the quadratic transformation of this variable was also significant, $\beta = 24.40, t = 3.51, p.0001$, suggesting a steep increase before the leap and a severe reduction after the jump. The recovery interval exhibited a decreasing trend in RMSSD, $\beta = 1.14, t = 1.85, p.072$. The time intervals accounted for 10% of the overall RMSSD variation.[1]

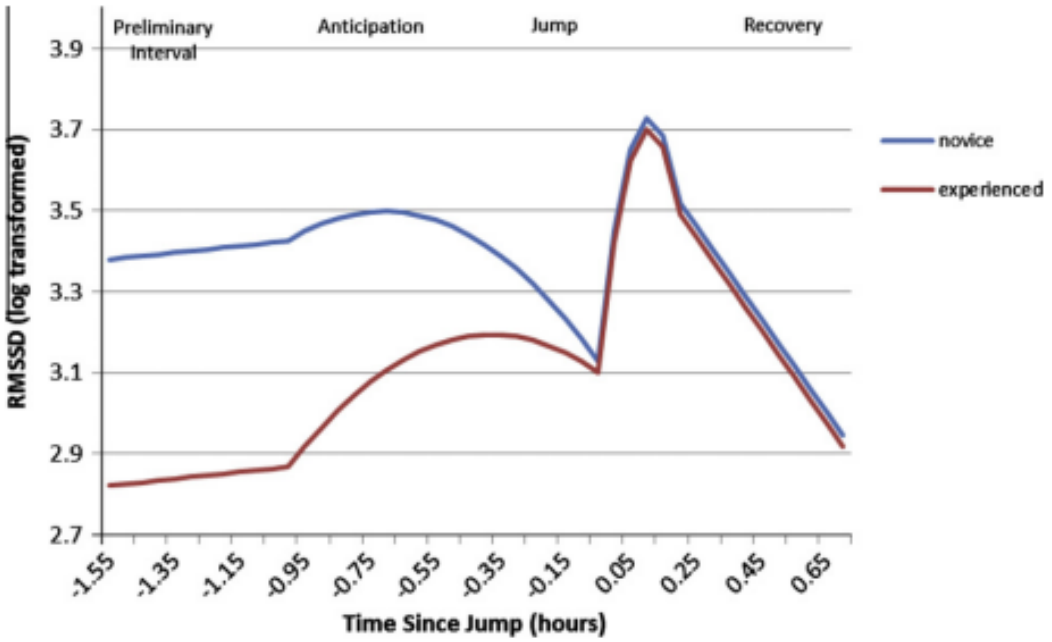

Figure 2: Before, during, and after jumping, RMSSD in novice and expert jumpers

4.3.1 Quality of Current Spacesuits

Layers of the Extravehicular Mobility Unit (EMU) The modern space suit is made up of 14 different layers of material, each of which contributes to the astronaut's survival in their own way. *The Liquid Cooling Ventilation Garment* is the three innermost layers make up the liquid cooling ventilation garment. This garment is made from a thin spandex layer that stays snug to the body and is designed to help cool the astronauts. This water comes from the astronaut's backpack or an umbilical cable leading to the ship, as well as recycled water from the astronaut's perspiration and water vapor in the carbon dioxide he or she exhales. The *bladder layer* is located above this layer and is perhaps the most important to the astronaut's life. The bladder layer maintains the necessary pressure dynamic to safeguard the astronaut by holding the same oxygen that the astronaut will breathe inside the limited space of the bladder layer. The role of the layer just above the bladder layer remains closely related to the importance of the bladder layer since this layer guarantees that the bladder layer keeps the appropriate structure for the astronaut. This layer, like the bladder layer of a spacesuit, is constructed of the same material as can be found in a tent and helps to keep the form of the tent. The following layer is particularly intended to withstand any tears that may develop, as direct exposure to space is extremely harmful. This layer is appropriately known as the *ripstop layer*. Following the ripstop layer are seven layers of *mylar insulation*, a substance commonly employed in food storage. This mylar insulation serves to keep the astronaut's temperature stable, much like a thermos or a cooler does for your food. The last layer is made up of three different fabrics, each with its specific function. One of the layers is made of Kevlar, the material used to build bulletproof vests, and acts as a general-purpose defensive layer. The other two layers are

intriguing because they perform virtually opposing functions: one is waterproof and the other is fire-resistant.

NASA's new suit, known as the Exploration Extravehicular Mobility Unit, or xEMU for short, will be used by astronauts on the Artemis lunar exploration mission. While the spacesuit is still in development, its features are complete. It is currently being tested underwater, with orbital testing expected for 2023. It can endure temperatures as low as **-250 degrees Fahrenheit** in the shade and as high as **250 degrees Fahrenheit** in the sun. The Life System Support backpack holds the suit's power and breathing air, as well as harmful gasses and temperature regulation. Because the electronics and plumbing systems are miniaturized to allow built-in duplicates, some system failures are less of a problem. It might potentially lengthen spacewalks. The headgear contains voice-activated microphones that detect astronauts' voices when they speak. A tiny foam block in the headpiece allows an astronaut to scratch his nose. The gloves have heaters in them to keep their fingertips toasty while still providing dexterity. The suit's flexible components are constructed of up to 16 layers of material that serve diverse roles. The suit is made of interchangeable pieces and might provide an improved habitat for the Martian.

4.4 Possible Space Diving Issues and Atmospheric Re-entry

Problems during landing on earth from space

- Accelerating enough speed to fall toward the earth
- Protecting the jumper against the heat of 2,500°F (1,370°C)
- Avoiding other objects, such as meteoroids and satellites
- Decelerating the speed of the jumper before deploying the parachute

4.4.1 Parachute Opening Shock and Impact Forces

One of the potential problems with space diving is parachute opening shock. Though a low altitude opening at typical terminal velocity would most likely result in minor shock, as altitude and free fall speed rise, so does shock. Parks College Parachute Research Group conducted relatively few experiments on parachute opening shock for the USPA (United States Parachute Association) [Potvin, Peek, 2013]. According to anecdotal testimony from special forces operators and high altitude jumpers, apertures at higher altitudes are substantially "harder" than openings at lower altitudes. According to firsthand experiences, there is a clear relationship between altitude and the initial shock. The majority of parachutes open in 1 to 8 seconds. Opening causes a significant amount of drag, resulting in a quick deceleration. This rapid braking provides the jumper with instantaneous g-load, but it also swiftly diminishes. It happens so quickly that it seems like a blow to the gut. Potvin and Peek's studies for the USPA revealed that the majority of the tested parachutes provided an opening shock of 3 to 6 g [Potvin, Peek, 2013]. In other severe situations, the weight reached 12, and even 18 g [Potvin, Peek, 2013]. However, the study does not clarify how such measures were obtained. It does, however, important, as demonstrated in the Methodology portion of this study. The USPA study also suggests that good parachute packing practices can minimize the majority of difficulty opening incidents [Potvin, Peek, 2013]. However, careful packing may not be enough for space diving since a variety of factors impact the opening shock. Severe opening shock can cause spine and muscular injuries, as well as fractured bones [Burton, 1999]. It requires additional investigation in order to create novel mitigating techniques.

5 Methodology & Calculations

5.1 Considering the most ideal conditions

We can jump from the thermosphere. As there are lack of molecules, our *Akashchari* won't feel the heat. Actually, he will feel cold. But suit temperature will rise while jumping from the mesosphere.

So let's assume that the person has jumped from the Mesosphere(We will discuss in later sections why we didn't choose higher altitudes).

The temperature in the mesosphere ranges from -2.5 degrees Celsius to -90 degrees Celsius. As we have a spacesuit that can withstand a maximum of 250°F(121.11°C), we can think that the change in temperature of his spacesuit is approximately 121.11-25 = 96.11°C = 96.11K $\Delta\theta$.
So,

$$\frac{1}{2}mV_t^2 = mS\Delta\theta$$

$$\longrightarrow V_t = \sqrt{2S\Delta\theta}$$

$$\longrightarrow V_t = \sqrt{2 \times 1000 \times 96.11}$$

$$\longrightarrow V_t = 438.43m/s2$$

Again, to gain this velocity,

$$h = \frac{V_t^2}{2g}$$

$$\longrightarrow h = \frac{438.43^2}{2 \times 9.81}$$

$$\longrightarrow h = 9797.15m = 9.8km(approximately)$$

So this height is above Stratosphere range. So the total height must be $(80 + 9.8)km = 89.8km$
Now, in the next part, we'll look at whether the calculations are sound and whether jumping from height is feasible.

5.2 Free fall differential equation for Laplace's isothermal atmosphere

It can be presented as:

$$\frac{mdV}{dt} = -mg + K \times V^2 \times e^{-z/\lambda}$$

$$K = 0.5C_d\rho A$$

By substitution $dt = \frac{dz}{V}$, we get,

$$\frac{mVdV}{dz} = -mg + K \times V^2 \times e^{-z/\lambda}$$

$$\longrightarrow \frac{VdV}{dz} = -g[1 - \frac{K}{mg} \times V^2 \times e^{-z/\lambda}]$$

$$\longrightarrow \frac{VdV}{dz} = -g[1 - (\frac{V}{V_t})^2 \times e^{-z/\lambda}]$$

where , $V_t = \sqrt{\frac{mg}{K}} = \sqrt{\frac{mg}{0.5C_d\rho A}} = TerminalVelocity$
$g = g(h, \phi$
$C_d = C_d(Ma, Re)$
$\rho = \rho(h, T$

Term $K \times U^2 \times e^{\frac{-z}{\lambda}} = K \times \frac{V}{V_t^2}$ represents Drag forces impacting on objects.

$$G_g = \frac{V}{V_t})^2 \times e^{\frac{-z}{\lambda}}$$

G-Forces are expressed in a number of Earth's g. Finally,

$$\frac{V}{V_t} = a \times e^{\sqrt{-ae^{\frac{-z}{\lambda}}}} \times \sqrt{F(z)}$$

$$F(z) = \frac{z_0 - z}{\lambda} + \sum_{n=1}^{\infty} \frac{a^n \times (e^{\frac{-nz}{\lambda}} - e^{\frac{-nz_0}{\lambda}})}{nn!}$$

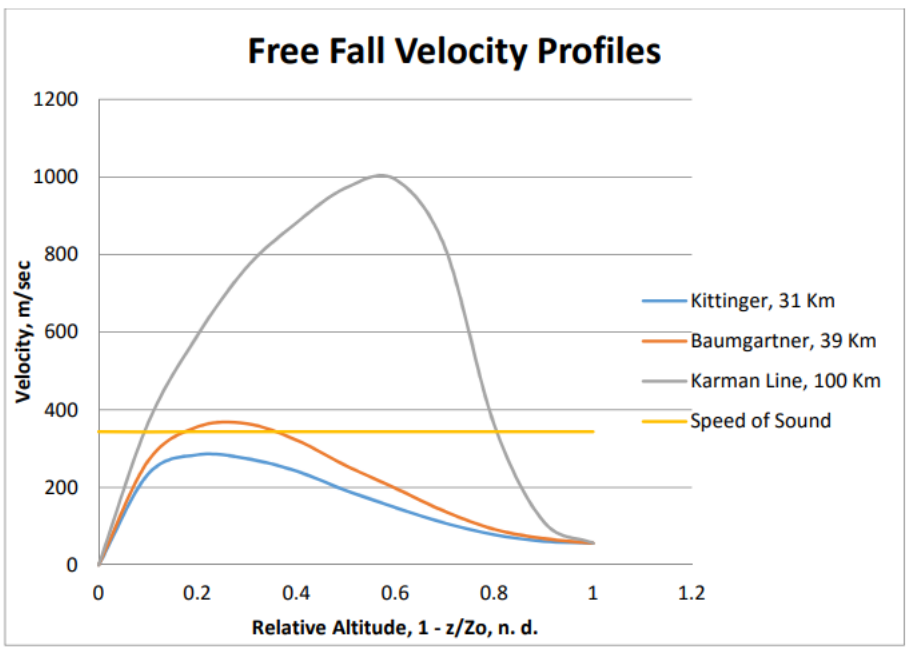

Figure 3: Profiles of free fall velocities for objects traveling in Laplace's (exponentially changing density) atmosphere are compared. (various starting elevations are represented by various colors. When the abscissa $(1 - \frac{z}{z_0})$ climbs from 0 to 1, the physical coordinate Z of a free falling object changes from Z_0 to 0 (zero). The sound speed of 343.2 m/s is shown on the graph for reference.)

Free fall velocity in stratospheric skydiving must be a great challenge, as they are substantially higher than free fall velocities in normal parachuting (J. Kittinger's parachute leap from altitude 31 Km serves as a reference jump here).

The analytic solution of U(z):

$$\frac{V}{V_t} = a \times e^{\sqrt{-ae^{\frac{-z}{\lambda}}}} \times \sqrt{F(z)}$$

$$F(z) = \frac{z_0 - z}{\lambda} + \sum_{n=1}^{\infty} \frac{a^n \times (e^{\frac{-nz}{\lambda}} - e^{\frac{-nz_0}{\lambda}})}{nn!}$$

where $a = \frac{2g\lambda}{V_t^2} = 46.686$

5.3 Drag Force Expression

$$G_g = \frac{V}{V_t})^2 \times e^{\frac{-z}{\lambda}} = a^n \times (e^{\frac{-z}{\lambda}} - \frac{-z}{\lambda}) \times F(z)$$

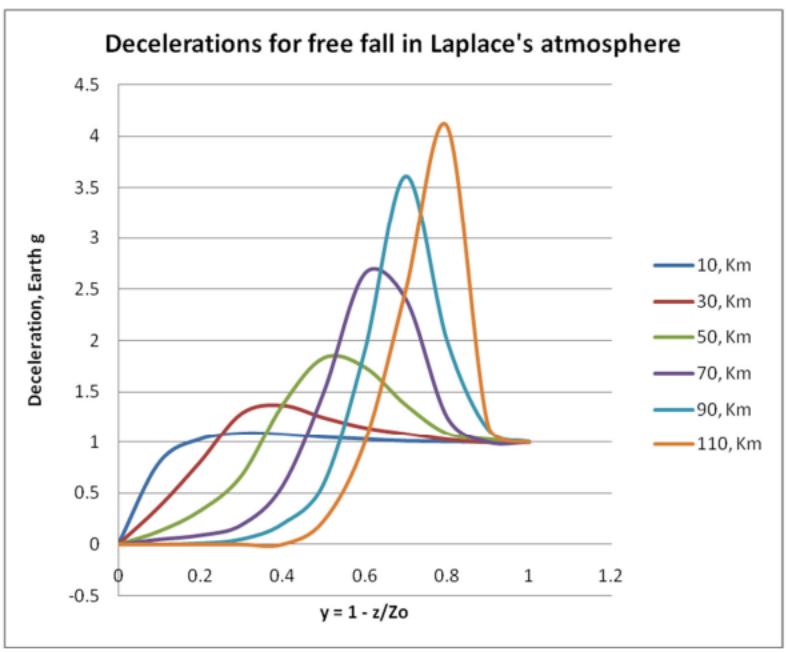

Figure 4: Profiles of free fall velocities for objects traveling in Laplace's (exponentially changing density) atmosphere are compared. (various starting elevations are represented by various colors. When the abscissa $(1 - \frac{z}{z_0})$ climbs from 0 to 1, the physical coordinate Z of a free falling object changes from Z_0 to 0 (zero). The sound speed of 343.2 m/s is shown on the graph for reference.)

According to these calculations, even without a parachute, a skydiver must endure G-Forces of 4 Earth's g or greater (in a horizontal oriented position, this level is manageable but requires specific training).

5.4 Analytical solutions for V_t

5.4.1 $V_t = constant$

$$\int_{V_0}^{V} dv/(1 - v^2/v_t^2) = 1/2 v_t ln[(v_t + v)(v_t - v_0)/(v_t - v)(v_t + v_0)] = g \int_0^t dt = gt$$

$$v_t = v_t \times [v_t + v_0 - (v_t - v_0)exp(-2gt/v_t)/v_t + v_0 + (v_t - v_0)exp(-2gt/v_t)]$$

On the other hand,

$$h(t) = h_0 - (1/2)gt^2 + (1/12)(g^3t^4/v_t^2) + ...$$

due to **gravity** & **1st order drag**.

Finally,

$$v(h) = v_t \times \sqrt{1 - (1 - v_0^2/v_t^2)exp[(2g/v_t^2)(h - h_0)]}$$

5.4.2 $V_t \neq constant$

$$\rho = \rho_0 \times exp((-h - h_0)/H)$$

$$H(T) = TR_s/g_0 = T \cdot 29.28mK^-1$$

finally,

$$\Delta v = g(1 - v_0^2/v_t^2) \times \Delta t - (v_0 g^2/v_{t_0}^2)[1 - v_0/v_{t_0}^2 + v_0/2gH] \cdot \Delta t^2$$

5.5 Calculations for parachute opening shock

5.5.1 Parachute opening in uniform atmosphere conditions

$$\frac{mVdV}{dz} = -mg + K \times V^2$$

$$\longrightarrow \frac{VdV}{dz} = -g[1 - \frac{K}{mg} \times V^2]$$

$$\longrightarrow \frac{VdV}{dz} = -g[1 - (\frac{V}{V})^2]$$

From this, we can find,

$$F = \frac{\frac{V}{V_t} - 1}{\frac{V}{V_t} + 1}$$

5.5.2 Evaluation of the first G-force shock at various altitudes beneath the Kármán Line

$$\frac{\Delta V}{\Delta t} = g \times ((\frac{V}{V_t})^2 - 1)$$

$$G = \frac{1}{g} \times \frac{\Delta V}{\Delta t} = (\frac{V}{V_t})^2 - 1)$$

In the last phrase, this is the purpose of evaluation for Vt. $V_t = 16.8m/s$ is the result of calculations based on preliminary parachute jumps in the atmosphere at sea level. This number appears to be higher than direct measurements of terminal velocities after parachute opening ranging from 3.7 to 7.1 m/s, but this is due to not taking account into atmospheric density changes with altitude and provides average estimates for Vt over the entire range of altitudes between 0 and 100 Km.

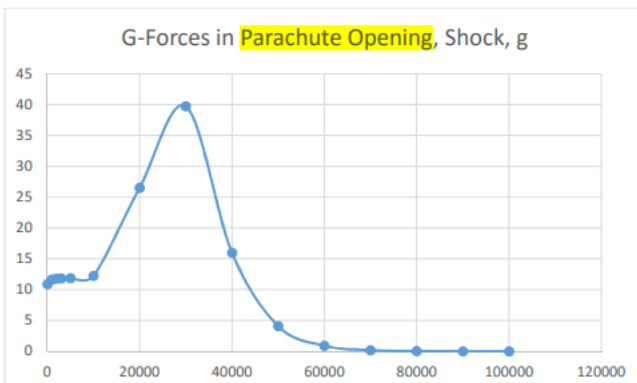

Figure 5: Shock of Parachute Opening at Various Altitudes

We can clearly see from the preceding calculations that all of the equations work for value beneath the Karman Line. Because our result is less than 100km, we may declare that 89.8km is a valid result.

6 Discussion

We have found all the according to present conditions. But we can make future spacesuits with special fabrics, which can be called **Future Suits**. Shape memory alloy coils are essentially springs that return to their original unstretched shape when they are heated. The pressure created by using these shape memory coils in tandem in a cuff matched the pressure needed to support a human in space.

Modern space suits achieve this pressure by essentially putting the in a gaseous chamber but these future space suits would achieve this pressure by mechanically squeezing around the astronaut. This means that these future space suits would be incredibly less bulky than the space suits of the present day.

Another unique feature of a space suit composed of shape memory alloys would be that it could squeeze around an astronaut when it reverts to its original shape and then when cooled it would stretch and allow the astronaut to exit.

7 Conclusion

References

[1] Vadim Rygalov Denys Bulikhov. Stratospheric sky-diving: parachute opening shock and impact forces analysis. 2015.